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The Pentium 4 3.2 GHz Extreme Edition (EE) is Intel's latest release to counter the Athlon 64 FX threat. Apart from a whopping 2 MB on-die L3 cache on the Pentium 4 EE, it isn't too much of a departure from the regular Pentium 4 3.2 GHz. The FSB is the same, and so is the core architecture. The EE doesn't have any extra processing units, and depends solely on the tri-level cache architecture.

It has a 2 MB L3 cache on top of the 8 KB L1 cache and the 512 KB L2 cache that was introduced with the Northwood core. This increases the the transistor count, which ultimately increases power consumption. The L3 cache does push up the EE's scores, but lets the processor down in multi-tasking jobs and video encoding. Overall, it doesn't do too much for the EE.

The 64 FX is based on a spanking new core codenamed Hammer. The L2 cache has been increased to 1 MB. Why is AMD doing this? Because the current x86 family of processors is limited to addressing only 4 GB of memory. Sooner or later, make no mistake, this limit will need to be crossed.

AMD has redesigned the core for this CPU. Part of the older 32-bit design is still in place, but registers have been added that are active in the 64-bit mode. The term '64-bit' that is bandied around does not mean a giant leap, just that the data path has been changed from 32-bit to 64-bit.

The 64 FX also includes some on-die features that makes it perform better than the Barton core: HyperTransport links, the Silicon On Insulator (SOI) technology and SSE2 support. HyperTransport is a high performance, point-to-point link for integrated circuits. The memory controller is integrated on the processor, the CPU can communicate directly with memory without depending on the motherboard chipset.

This means the FSB speed doesn't matter, and AMD claims that the 64 FX can communicate with motherboard chipsets at speeds of up to 1,600 MHz. The memory controller interfaces with system RAM through a DRAM controller, and can be modified as new memory standards emerge. One drawback of the 64 FX is the need for registered or ECC memory—ECC memory comes at a premium.

Conclusion

Intel has a tough task ahead—making a processor to combat the 64 FX.

In all the tests that we ran, the 64 FX dominated the EE, except in the SiSoft Sandra 2004 and PCMark PRO tests. We also carried out tests for the 64-bit capabilities of the 64 FX in Red Hat Enterprise Linux WS. Our suite of Linux 64-bit tests included Unix Bench and nbench. The Athlon FX performed better than the Pentium 4 EE in most of the tests. This goes to show that the processor does well in large double-precision floating-point calculations. The gaming tests were also dominated by the 64 FX, except for the *Quake III* test run, where the EE made the Athlon look puny.

The Pentium 4 has its strong points in being compatible with current motherboards and DDR RAM modules, whereas AMD has gone through a complete makeover for the 64 FX. The EE's gaming scores were commendable, however, and one can't dismiss it offhand.

Microsoft is planning a 64-bit Windows XP release in a few months' time, and there are already 64-bit Linux OSes. *Unreal Tournament 2004*, to be out late this year, is touted as the first game supporting 64-bit processors. Overall, the Athlon 64 FX-51 processor is currently the one that we recommend, if you have that kind of money and if you want to have the 64-bit tag on your desktop, with performance across all applications.

The EE, on the other hand, seems to be an Emergency Edition from Intel, with not much performance gain over the regular Pentium 4 3.2 GHz. The added L3 cache does not give it much of a boost in running applications any faster. Intel needs to bring out something new to combat the Athlon FX. Perhaps the Prescott will bring in good news for them!

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